COGNIFY DEMO SYLLABUS

Program: B.Tech Computer Science / Information Technology
Semester: 3
Document Type: Sample syllabus for PDF extraction demonstration

Note: This is a simplified fictional/demo syllabus for testing Cognify. It is not an official AKTU syllabus.

Subject 1: Data Structures

Unit 1: Introduction and Arrays

- Topic 1: Data structures and classifications
- Topic 2: Time complexity and asymptotic notation
- Topic 3: Arrays and memory representation
- Topic 4: Searching: linear search and binary search

Unit 2: Stacks and Queues

- Topic 1: Stack operations and applications
- Topic 2: Infix to postfix conversion
- Topic 3: Queue and circular queue
- Topic 4: Recursion and Tower of Hanoi

Unit 3: Linked Lists

- Topic 1: Singly linked list operations
- Topic 2: Doubly linked list
- Topic 3: Circular linked list
- Topic 4: Polynomial representation using linked lists

Subject 2: Computer Organization and Architecture

Unit 1: Computer Organization

- Topic 1: System bus structure
- Topic 2: Bus arbitration
- Topic 3: Processor organization
- Topic 4: Addressing modes

Unit 2: Computer Arithmetic

- Topic 1: Carry look-ahead adder
- Topic 2: Booth multiplication algorithm
- Topic 3: Restoring division
- Topic 4: Floating point representation

Unit 3: Instruction Processing

- Topic 1: Instruction formats
- Topic 2: Instruction cycle
- Topic 3: Micro-operations
- Topic 4: RISC and CISC architectures

Subject 3: Digital Electronics

Unit 1: Number Systems and Logic Gates

- Topic 1: Binary, octal and hexadecimal numbers
- Topic 2: Boolean algebra
- Topic 3: De Morgan laws
- Topic 4: Basic logic gates

Unit 2: Combinational Circuits

- Topic 1: Karnaugh maps
- Topic 2: Half adder and full adder
- Topic 3: Multiplexers and demultiplexers
- Topic 4: Encoders and decoders

Unit 3: Sequential Circuits

- Topic 1: SR flip-flop
- Topic 2: JK flip-flop
- Topic 3: D and T flip-flops
- Topic 4: Counters and registers

Subject 4: Discrete Structures and Theory of Logic

Unit 1: Logic and Sets

- Topic 1: Propositions and logical connectives
- Topic 2: Truth tables
- Topic 3: Set operations
- Topic 4: Relations and functions

Unit 2: Combinatorics

- Topic 1: Permutations and combinations
- Topic 2: Pigeonhole principle
- Topic 3: Mathematical induction
- Topic 4: Recurrence relations

Unit 3: Graph Theory

- Topic 1: Graph terminology
- Topic 2: Paths and cycles
- Topic 3: Trees and spanning trees
- Topic 4: Graph traversal basics

Subject 5: Python Programming

Unit 1: Python Fundamentals

- Topic 1: Python syntax and variables
- Topic 2: Data types and operators
- Topic 3: Conditional statements
- Topic 4: Loops

Unit 2: Collections and Functions

- Topic 1: Lists and tuples
- Topic 2: Dictionaries and sets
- Topic 3: Functions and parameters
- Topic 4: Recursion

Unit 3: Files and Exceptions

- Topic 1: String operations
- Topic 2: File handling
- Topic 3: Exception handling
- Topic 4: Modules and packages